Weekly Record 5

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Replenishment, controls and performance
Carrefour and Magnum Cash&Carry

1 Concept

Operations management coordinates people, materials and information to deliver a useful service at an acceptable cost. Resource planning balances availability against the cost of holding stock. A process map shows where work passes between people and where a delay or error can affect the whole service [1].

Concept Check, section 10.4: how is technology used in resource planning? It can combine sales and stock data to support purchasing. The benefit depends on accurate inputs and workable decisions. Faster calculations alone will not fill an empty shelf, and a forecast should not become a purchase without review.

2 Application

The process is replenishment of one product category in one Magnum store. Its retail activities and product acceptance rules establish the context [14], [22]. The current-state sequence below is an evidence-informed baseline to confirm with staff; it is not an observed account of Magnum's internal procedures. The proposed bottleneck is reconciling stock and promotion information before deciding quantities.

Step	Current state to confirm	Proposed future state
1	Check sales and stock.	Automated: validate sales and stock inputs; block failed checks.
2	Estimate demand and order quantity using the existing method.	AI Executes with Human Review: forecast demand and draft quantities.
3	An authorised manager approves the order.	Human + AI: staff interpret promotions and supplier constraints. Human Only: the category manager checks exceptions and approves or changes the order.
4	Send the order and receive goods.	Send only the approved order and receive goods.
5	Check shelves and record results.	Check shelf availability and costs. Automated logs preserve inputs, revisions and approvals for review.

If records disagree or a forecast fails, staff stop the affected recommendation and alert the category manager. Unresolved supply problems go to the supply-chain lead; technical faults go to IT. Staff prepare a manual order from verified sales and a physical stock count. The category manager approves it and authorises any restart after correction. No unapproved order is sent [16].

The following six-month pilot targets are proposals, not Magnum results. Establish a baseline before starting, use the same products and checking schedule, and compare results with the existing ordering method.

Measure	Definition and proposed target
Shelf availability	Available product checks ÷ all scheduled checks × 100. Target: at least 98%.
Forecast error (WAPE)	Sum of absolute forecast errors ÷ sum of actual units sold × 100. Target: a 10% relative reduction from baseline.
Cost and waste checks	Proposed reductions of 10% each in expired/damaged stock value and staff cost per order. Include software, training and correction costs when assessing net benefit.

Lost sales during stockouts can conceal demand, so WAPE must be interpreted alongside shelf availability. The category manager should review exceptions daily and trends weekly, using store feedback to explain unusual results [1].

Carrefour's documented forecasting initiative makes PILOT a defensible transfer decision [20]. Week 3's stock-data problem remains a stop condition. Expand only if availability reaches the target, forecast error improves, net benefit remains positive and staff can use the fallback. This map provides the operations argument and on-screen evidence for Assignment 2.

References

Source numbers and full links are in the cumulative Evidence Folder.

[1] L. J. Gitman et al., Introduction to Business, 2nd ed. OpenStax, 2026, secs. 6.5 and 10.1, 10.4–10.5.

[14] Magnum Cash&Carry, "About us," n.d.

[16] NIST, "Artificial Intelligence Risk Management Framework AI RMF 1.0," NIST AI 100-1, Jan. 2023.

[20] SAS, "Multinational Carrefour relies on SAS AI to optimize supply chain management and reduce food waste," Jan. 14, 2019.

[22] Magnum Cash&Carry, "Corporate rules," n.d.